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Sustainable Software Engineering - Introduction to Software Sustainability



Gemma Catolino | 2026-2027

Sustainable Software Engineering: What is Sustainability?

Sustainability is a widely used term and refers to the capacity of something to last a long time.

SOME DEFINITIONS:

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According to the UN, sustainable development needs to satisfy the requirements of three dimensions, which are society, the economy, and the environment.

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Sustainable Software Engineering: Three Dimensions

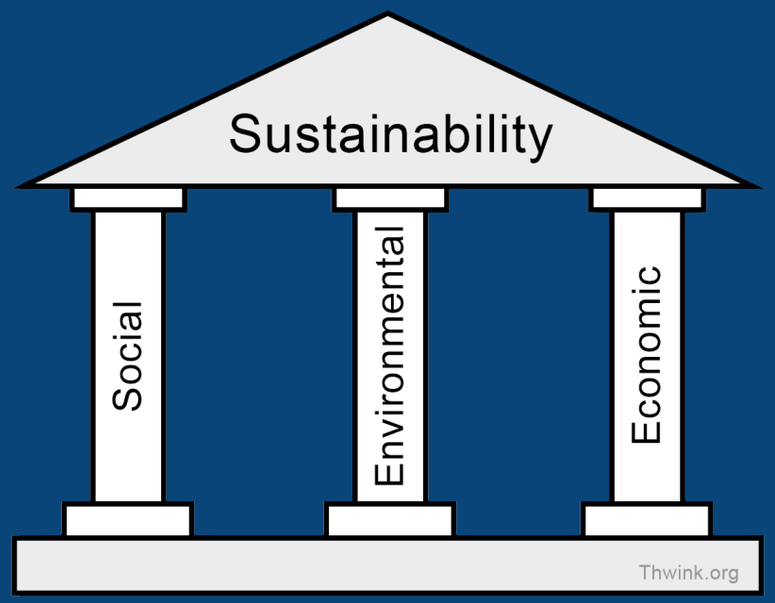
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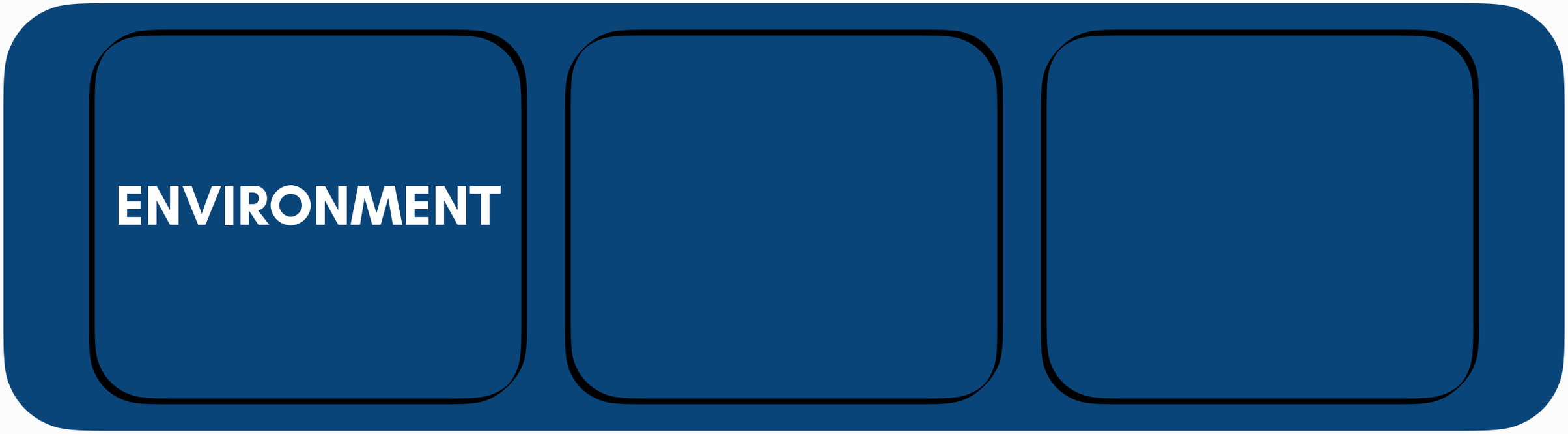
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Ensures that the environment is able to replenish itself at a faster rate than it is destroyed by human actions.

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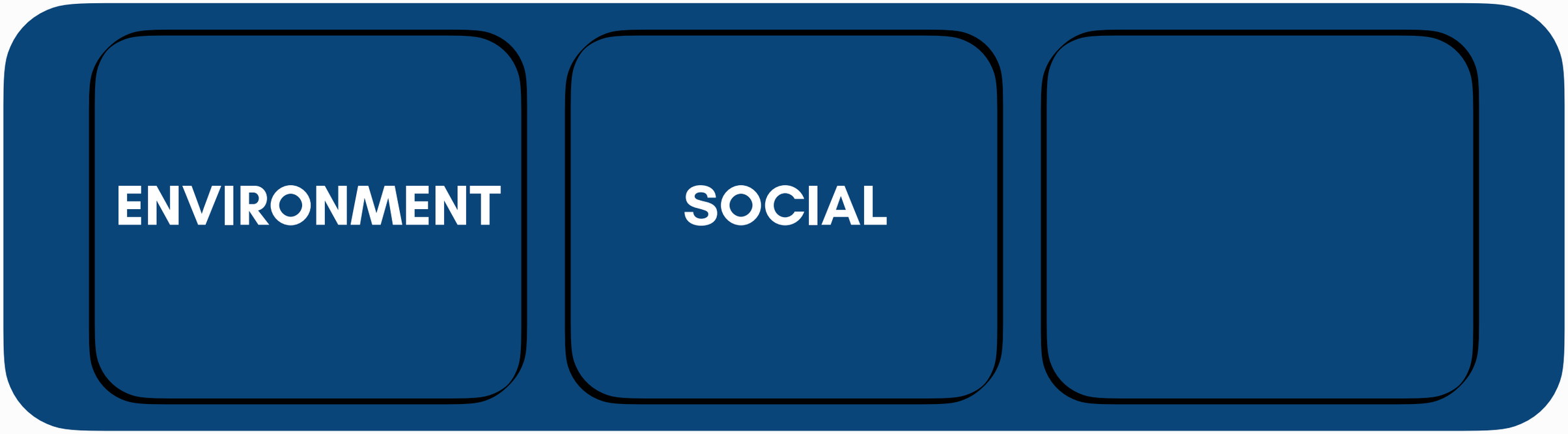
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ENVIRONMENT

Ensure creating a sustainable society which includes all actions that promote social equity and ethical consumerism.

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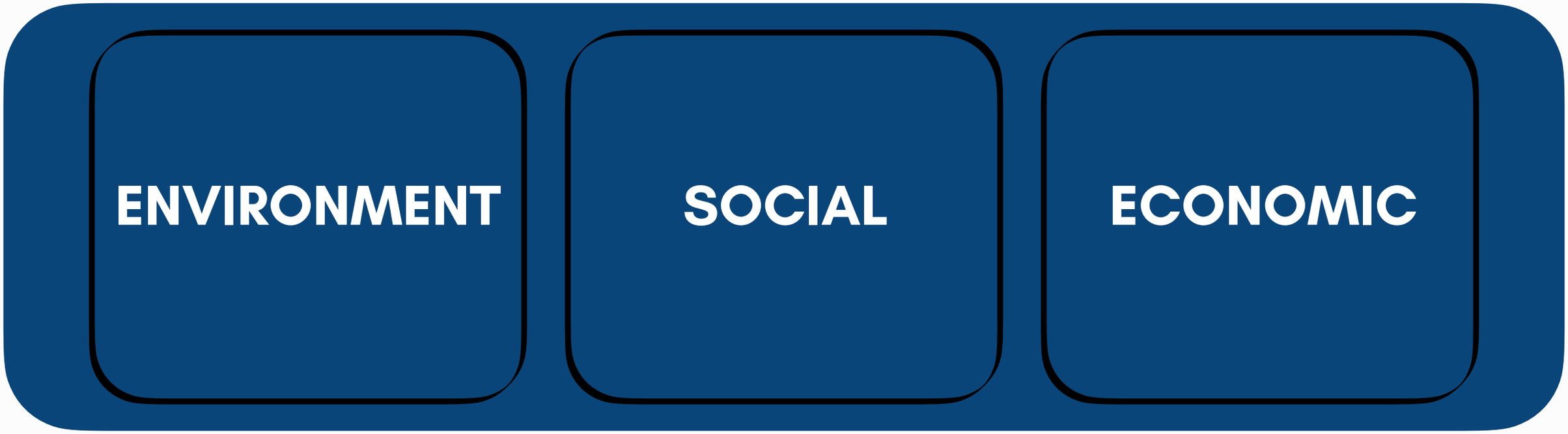
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SOCIAL

Ensures that our economic growth maintains a healthy balance with our ecosystem; it integrates environmental and social concerns into business.

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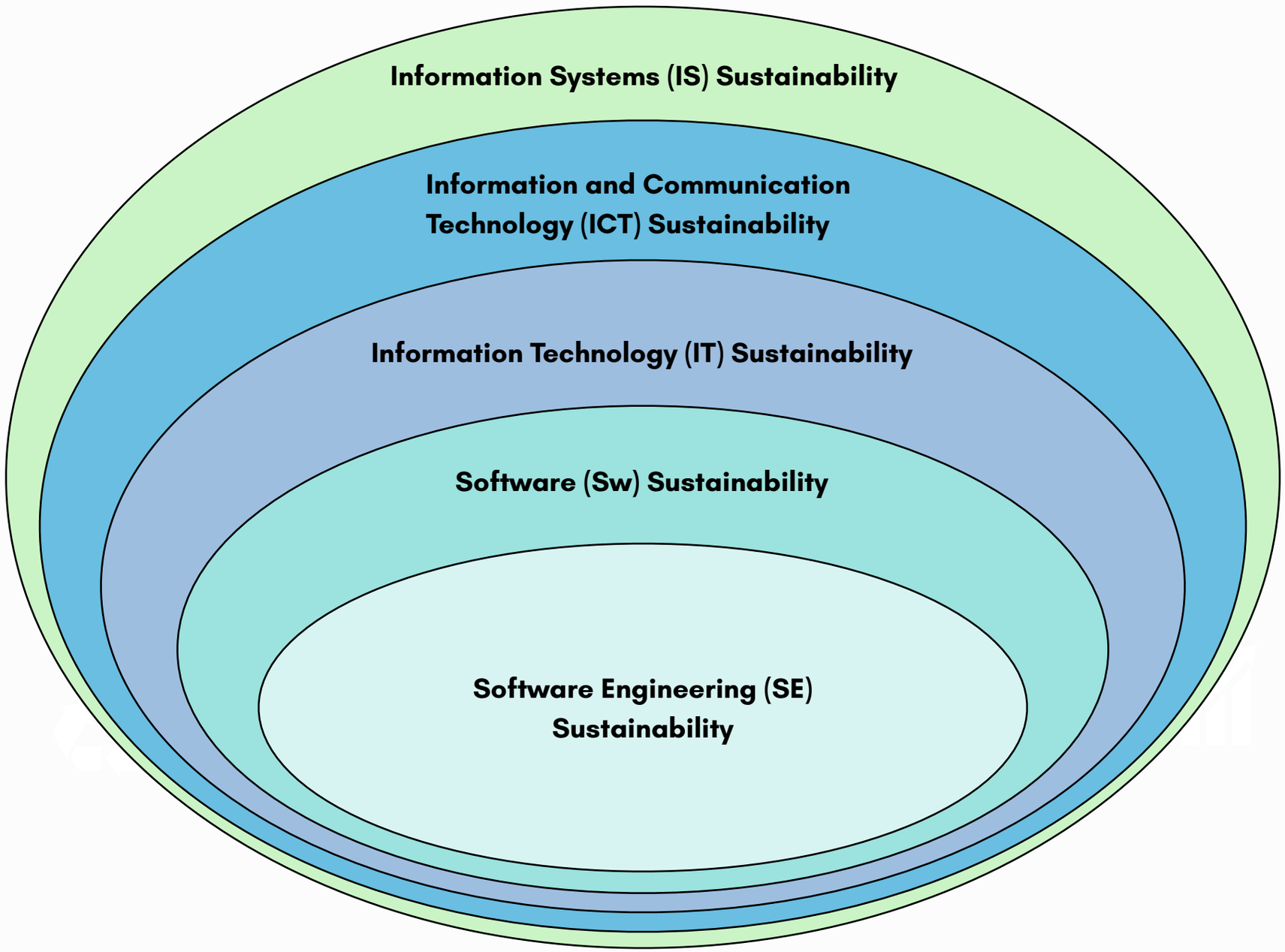


Developing energy-efficient cloud computing solutions helps companies reduce operational costs.

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Sustainable Software Engineering: Sustainability Levels

Another sustainability-related aspect discussed in the literature is the domain of its application.



SUSTAINABILITY LEVELS

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Information Systems (IS) Sustainability
Information Systems can play a central role in creating an ecologically sustainable society because of decades of experience in designing, deploying, and managing systems that solve complex problems.
IS sustainability emphasizes process change (analysis, measurement, improvement) to build environmentally sustainable organizations and societies.

Information and Communication Technology (ICT) Sustainability
Sustainable ICT aligns all processes with the principles of reduce, reuse, and recycle, and finds innovative ways to deliver sustainability benefits across enterprises. It includes reducing or eliminating materials, improving efficiency, and enabling greener practices in different sectors through ICT.

Information Technology (IT) Sustainability
IT sustainability is often understood as a shorthand for global environmental sustainability. It envisions an Earth where essential processes persist for long periods by reducing energy use, minimizing environmental degradation, and developing IT solutions that support a sustainable future.

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Level	Focus	Scope	Example
IS Sustainability	Processes & systems	How IS design, management, and process change make organizations more sustainable	Green Business Process Management to reduce waste
ICT Sustainability	Infrastructure & practices	Aligns ICT operations with Reduce-Reuse-Recycle; delivers sustainability benefits across enterprises	Cloud providers reusing servers, recycling e-waste, cutting network energy
IT Sustainability	Global environmental impact	IT as part of global sustainability: reducing energy/material use and enabling a sustainable future	Energy-aware algorithms, smart grids, eco-friendly IT solutions

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Software (Sw) Sustainability

The term Sustainable Software can be interpreted in two ways: (1) the software code being sustainable, agnostic of purpose, or (2) the software purpose being to support sustainability goals. Therefore, in this context, sustainable software is energy-efficient, minimizes the environmental impact of the processes it supports, and has a positive impact on social and/or economic sustainability.

Sustainable software is *"software, whose impacts on economy, society, human beings, and environment that result from development, deployment, and usage of the software are minimal and/or which have a positive effect on sustainable development."*

Software Engineering (SE) Sustainability

Sustainable software engineering integrates sustainability principles into the development lifecycle. It involves creating reliable, long-lasting software that balances user needs with reduced economic, social, and environmental impacts. Key practices include continuously assessing and optimizing sustainability effects across the entire lifecycle.

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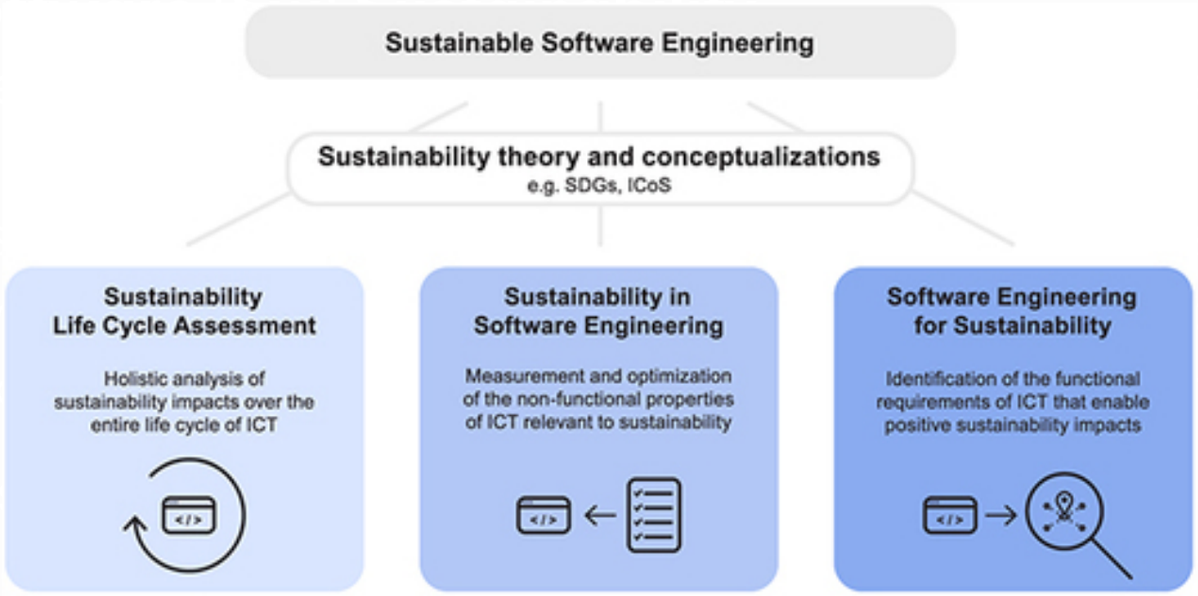
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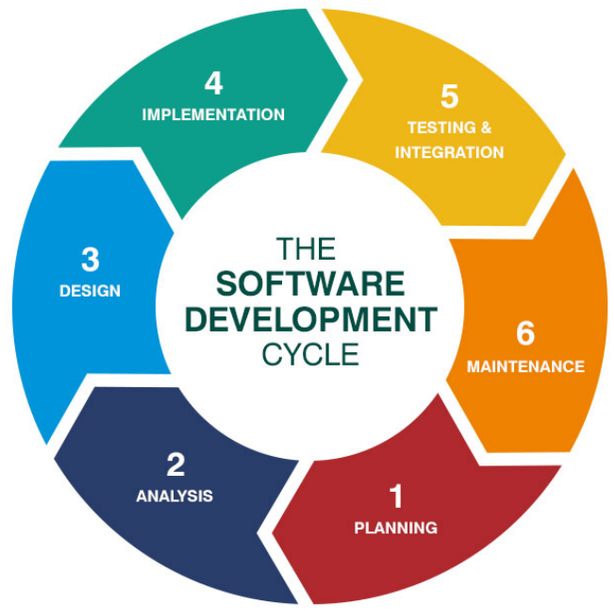
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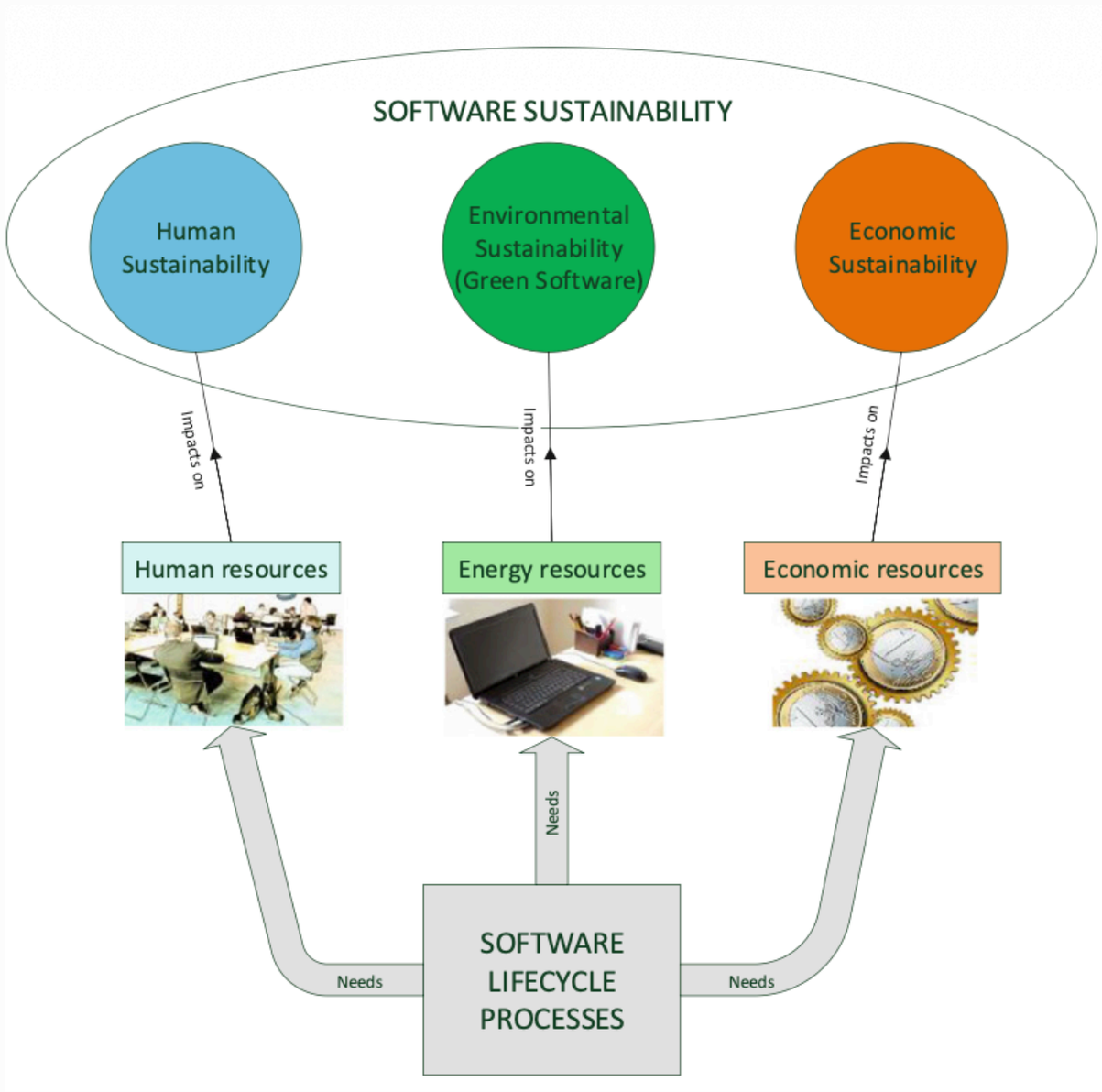




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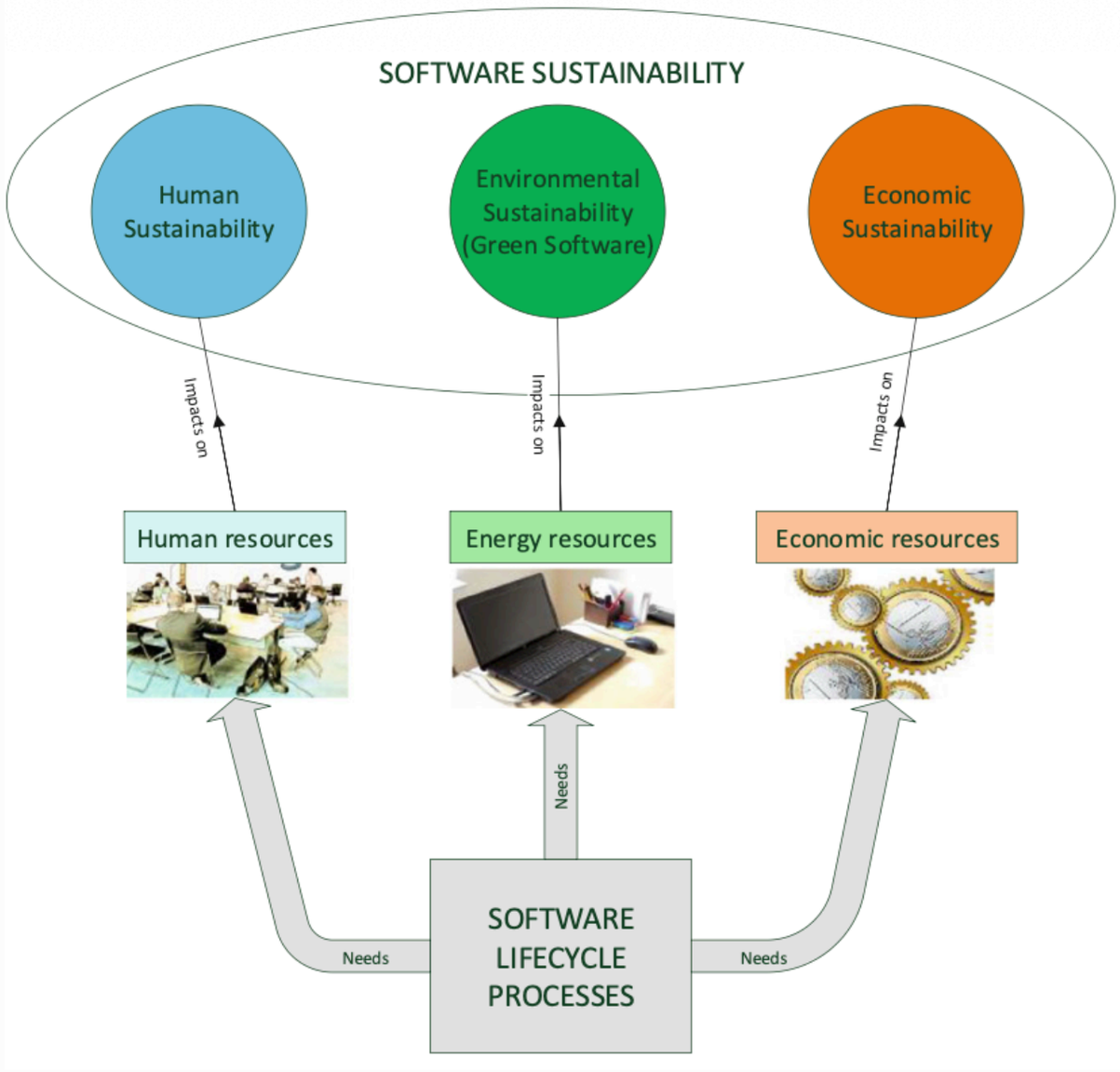
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Human sustainability

How software development and maintenance affect the sociological and psychological aspects of the software development community and its members. This encompasses topics such as labour rights, psychological health, social support, social equity, and liveability.

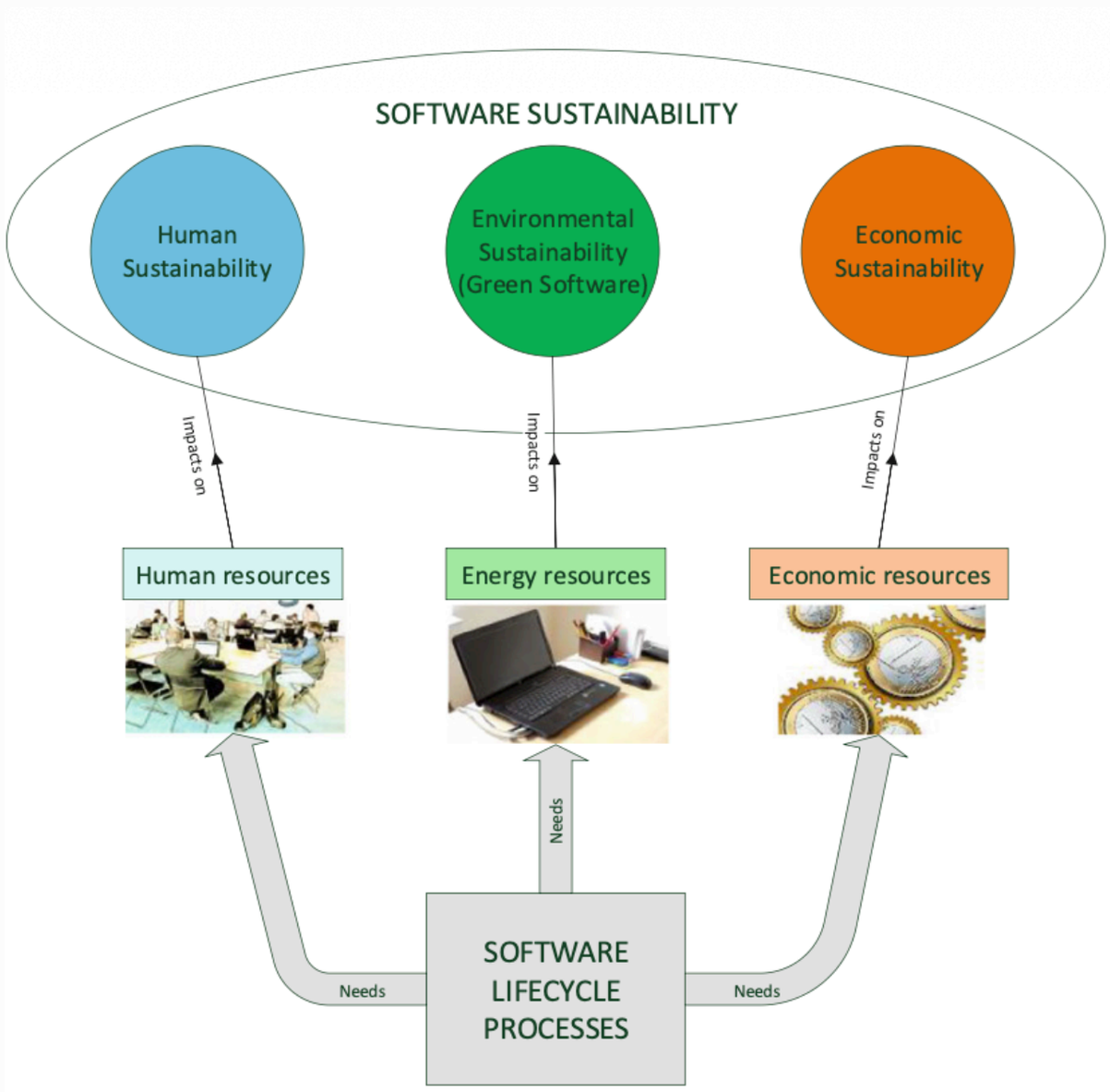
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Economic sustainability

How the software lifecycle processes protect stakeholders' investments, ensure benefits, reduce risks, and maintain assets.

Sustainable Software Engineering: Software Engineering (SE) Sustainability



Environmental sustainability

How software product development, maintenance, and use affect energy consumption and the usage of other resources.

Environmental sustainability is directly related to a software product characteristic that we call *"Green Software."*

Sustainable Software Engineering: Sustainable Development Goal (SDG)

The 2030 Agenda for Sustainable Development, adopted in 2015 by all United Nations Member States, provides a shared blueprint for peace and prosperity for people and the planet, now and into the future. At its heart are the 17 Sustainable Development Goals, which are an urgent call for action by all countries—developed and developing—in a global partnership.



https://sdgs.un.org/goals

Sustainable Software Engineering: SDG and Software Sustainability Dimensions

Goal 7: AFFORDABLE AND CLEAN ENERGY. Among the objectives of this goal one can find the improvement of effciency, which can be achieved through sustainable software, as pursued by the environmental dimension.

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Goal 9: INDUSTRY, INNOVATION, AND INFRASTRUCTURE. In this goal the need for inclusive and sustainable industrial development is mentioned, as is the need for technological solutions to ensure environmentally sound industrialization, plus the technological progress required to achieve environmental objectives, such as energy efficiency.
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Goal 12: RESPONSIBLE CONSUMPTION AND PRODUCTION. This goal promotes, among other things, resource and energy efficiency, green and decent jobs, and a better quality of life for all, increasing net welfare gains from economic activities by reducing resource use, degradation, and pollution through-out the whole lifecycle, while improving the quality of life.
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Goal 17: PARTNERSHIPS. These inclusive partnerships are built upon principles and values, a shared vision, and common goals that place people and the planet at the center. They also have to do with long-term investments in the information and communications technologies sector. This goal is thus related to the economic and the environmental dimensions.

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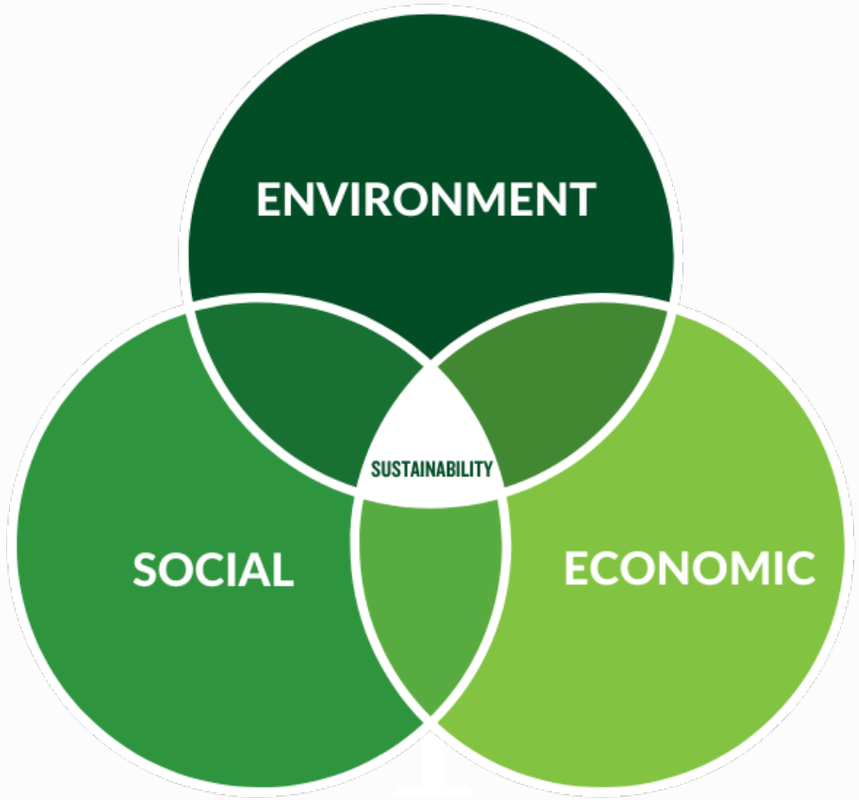
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Sustainable Software Engineering: SDG and Sustainable Software Engineering

SDG	Goal	Relationship with SSE	SSE Practices / Examples
1	No Poverty	Digital finance, e-governance, and social apps can reduce poverty.	Lightweight apps for low-resource devices, open-source financial software.
2	Zero Hunger	AI and data-driven agriculture rely on sustainable software.	IoT-enabled smart farming, efficient supply chain management software.
3	Good Health and Well-being	Reliable, secure software is key for healthcare.	Telemedicine platforms, secure health data systems.
4	Quality Education	Digital learning depends on scalable, inclusive infrastructure.	Open-source e-learning systems, offline learning apps.
5	Gender Equality	Inclusive design and bias-aware systems support equality.	Bias testing in ML models, inclusive UI/UX, diversity in teams.
6	Clean Water and Sanitation	Water monitoring and sanitation systems need efficient code.	IoT water sensors, real-time monitoring dashboards.
7	Affordable and Clean Energy	Software can cut energy use and optimize renewable grids.	Energy-aware coding, green cloud deployment.
8	Decent Work and Economic Growth	SSE fosters innovation while reducing burnout.	DevOps automation, fair open-source contributions.
9	Industry, Innovation, and Infrastructure	Sustainable digital systems enable resilient industries.	Efficient cloud services, containerization to reduce overhead.
10	Reduced Inequalities	SSE reduces digital divides with accessible tech.	Assistive technologies, multilingual/localized software, inclusive AI.
11	Sustainable Cities and Communities	Smart city apps need scalable and efficient code.	Traffic optimization algorithms, sustainable mobility apps.
12	Responsible Consumption and Production	SSE reduces e-waste and computing waste.	Modular software reuse, green coding standards.
13	Climate Action	SSE reduces carbon footprints directly and indirectly.	Carbon-aware software (e.g., CodeCarbon), low-power AI models.
14	Life Below Water	Ocean monitoring relies on efficient computing.	Satellite data analysis tools, pollution detection models.
15	Life on Land	Biodiversity tracking and conservation software need SSE.	AI for deforestation alerts, efficient GIS tools.
16	Peace, Justice, and Strong Institutions	Ethical and secure software fosters trust.	Blockchain for transparency, secure e-governance systems.
17	Partnerships for the Goals	Open collaboration accelerates sustainable solutions.	Open-source green software, global data-sharing platforms.

Sustainable Software Engineering: Learn by doing

Energy efficiency
Green computing
Carbon footprint
Resource optimization
Lifecycle management
Circular economy
Eco-design
Software longevity
Cloud sustainability
Green AI
Low-power computing
Performance per watt
Data center efficiency
Hardware co-optimization
Sustainable DevOps
Renewable energy integration
Social responsibility
Digital inclusion
Accessibility
Diversity & equity
Governance & compliance
Knowledge sharing



Community engagement
Fair labor practices
Developer well-being
Work-life balance
Resilience (social & technical)
Continuous improvement
Long-term maintainability
Technical debt reduction
Cost efficiency
Total cost of ownership
Return on investment (ROI)
Business value alignment
Productivity improvement
Economic resilience
Scalability
Innovation sustainability
Market competitiveness
Risk management
Reduced downtime
Value-driven design
Lifecycle cost optimization
Ethical AI

WHICH SUSTAINABLE DEVELOPMENT GOAL DID YOUR PROJECT ADDRESS?

Step 1 – Recall your project

Think of a **previous software project** (coursework, internship, hackathon, thesis, etc.).

Step 2 – Connect it to the SDGs

Look at the **17 UN Sustainable Development Goals**

Ask yourself:

- Which SDG(s) does my project contribute to?
- How does it help *directly* or *indirectly*?
- Which **dimension of sustainability** does it impact most?

Step 3 – Share & Discuss

Individually: Write down your chosen SDG(s).

Together: Compare and find common SDGs.

- Which SDGs are most/least addressed?
- Are all three sustainability dimensions considered?

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